

THE PROCESS
OF
GILDING AND BRONZING
PICTURE FRAMES

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PRELIMINARY REMARKS

ON

Gilding and Bronzing.

The art of gilding has been practiced from the earliest dates. Mention of it is made in the Scriptures, and it was known to the Egyptians, who made an extensive use of it in the ornamentation of their burial cases. It became to be generally used amongst the Romans soon after the downfall of Carthage, and was applied very profusely to the decoration of their dwellings and temples.

During the middle ages, and when the seats of learning and civilization were located in Italy and among the Moorish Arabs in Spain, the most elaborate use of it was made in almost every branch of art. It entered largely into architectural decoration, and the cathedrals and mosques were favorite objects for its display.

Among the Chinese, the use of gold leaf as a method of decoration has been practiced for an indefinite period, and its origin with them must be very ancient. Taken altogether, gilding is an art that has been and still is in more general use than any other; and the reason is, that it is almost always appropriate wherever called into requisition.

The art of gilding and of bronzing can readily be acquired by any one who is disposed to devote care and attention to it. It is an art having few, if any, amateurs. Those who practice it are generally those who follow it as a trade, and, perhaps, for the reason that such persons have but little time to spare from their profession, there is so little written information found on the subject.

The cyclopædias are exceedingly deficient in everything like a correct or reliable description of the art. The information which they give is mostly of an elementary character, and of a kind that is somewhat impracticable and very unavailable.

It was the want of a reliable work on the subject that suggested to me the publication of such a general exposition of the whole matter as would prove useful to those who might desire to be informed on it. I do not propose to do more than give a description of the various processes, and to endeavor to make such a description as simple and lucid as possible. It is no part of a gilders business to make the frame which he has to gild. Such used to be the case, but now there are factories where the frame is turned out completely made. There are a number of such factories in New York, and the person requiring a frame has only to select the desired pattern and have it made up according to the dimensions of the picture to be framed. Neither do I think it necessary to introduce the subject of preparing those materials which are made expressly for the purpose by regular dealers, and which, of course, will then be found more reliable than when made by inexperienced hands. There are several respectable dealers in New York who keep for sale almost everything connected with the gilding and bronzing processes.

At a time when every one is more or less interested in the subject of decorative art, and when many are seeking to qualify themselves towards proficiency in some one or other branch of it, perhaps no more appropriate subject could be offered to the public than that of gilding a picture-frame in all its parts.

No fine picture should be without a fine frame, and even with a fine frame an indifferent picture is very much improved; but taste and discernment are both required in the selection of such a frame as either a fine or an indifferent picture may require; and, although the artist in general is supposed to be the best qualified to determine as to the manner in which his painting should be framed, the frame-maker is really more competent for that duty, and for the best of all reasons—that it is his specialty.

Frames gilded in different colors of gold leaf are frequently seen, and, for some subjects, such frames may do very well; but, in general, the deep-colored gold is the most appropriate, and is the better suited to remain in favor with the public. There can hardly be anything richer than a frame finished in burnished and matted gold of a deep shade of color.

Bronze frames, which have recently come so much into vogue, are destined to retain their hold upon the public esteem for a long time. They are a separate style of decoration, one that is very rich, and which admits of great variety in the preparation, and they are particularly appropriate to many subjects, such as water colors and engravings.

The art of their preparation is comparatively new to this country, although it has been practiced for a long time in Europe; where, perhaps, the best work is yet done. It has hitherto been kept a profound secret by its votaries, and it has been a very difficult proceeding for an amateur to obtain the least insight into it. It is said that some bronzers, when practicing the art, shut themselves apart, in order to prevent their operations being watched. There is really no reason for so much secrecy on the part of those who have become possessed of the information concerning this process, for the process is, in fact, a very simple one. It does not require anything like the manipulation which gilding requires. It is rather more laborious, however, as the amount of burnishing of which it is susceptible is almost unlimited. This, however, is not labor lost, as it serves to enhance the attractiveness of the frame when finished.

It is the practice to use different colored bronze powders in combination, and these combinations are very rich in their effect. The silver, copper and verde-antique make rich contrasts. There is a great field for the exercise of taste in the bronzing process, and those who are adepts in it can command high prices for their work. The only drawback, and it is equally a drawback to many other excellent processes, is, that its popularity has been the means of causing a vast deal of cheap and inferior work to be brought before the public; but work that is really fine will continue to be regarded with favor wherever it may be found.



THE GILDING PROCESS.

I.

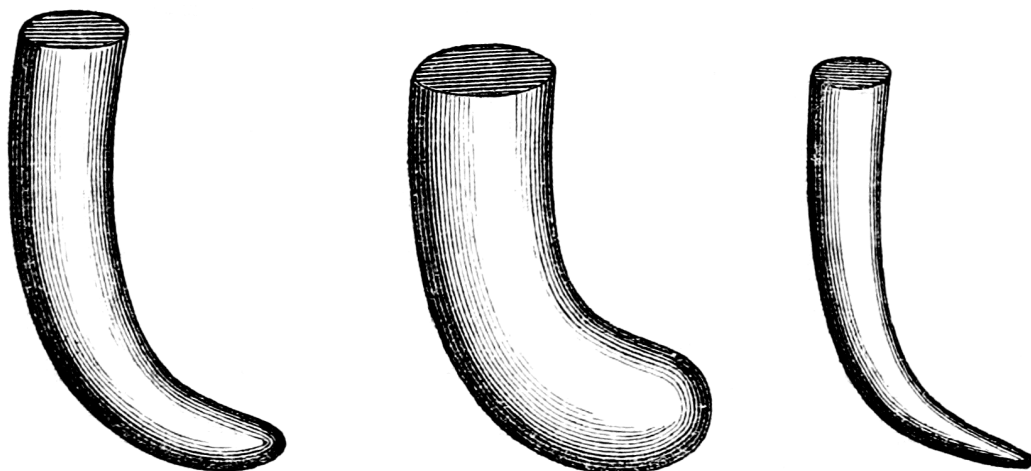
I will now proceed to give a list and description of the various implements and materials which are required in the process of gilding picture frames.

1.—A GILDING CUSHION, on which to lay the leaves of gold preparatory to cutting them into the sizes required for laying. This may be procured already made. It is made of a board one-half inch thick and nine and one-half inches long by five and one-half inches wide, covered with buckskin and padded with several thicknesses of Canton flannel. It usually has a shield of parchment, about four inches high, on one end, to protect the leaves from currents of air. Underneath, is a loop for the thumb, which is to be inserted in it when the cushion is to be held in the hand. There is also another loop to hold the gilding knife when not in use.

2.—A GILDING KNIFE, for cutting the gold leaf. It should be double-edged and have a tolerably keen edge; not so much so as to cut the cushion, but sufficiently so to divide the leaf without tearing it. When it becomes dull it may be readily sharpened by rubbing the edges with a piece of very fine sand paper.

3.—BRUSHES, both bristle and camel's hair. Numbers 4 and 5, of the bristle brushes, flat, are the sizes mostly required, either for applying the white coat or the oil-size. For the camel's hair brushes, numbers 4, 5 and 6 are very useful, as well as a number 6 or 8 lettering pencil (brush); also several fitch blenders, numbers 9 and 10 are good sizes; procure one of each.

4.—BURNISHERS. These must be of several shapes and sizes. Every gilder will become accustomed to some particular shape and size, and practice only will determine the most desirable one; but for ordinary use, the following *shapes* and *sizes* will be found sufficient for almost any work:



The burnishers should be made of flint, though agate is frequently used, particularly for burnishing silver leaf. They should be securely fastened in wooden handles by means of brass ferrules. These handles, as well as the stones, frequently become loosened from continual use and require refastening. This is done by renewing the filling with gum shellac or powdered rosin, and applying sufficient heat to set the stones. On cooling, they will be found to be again firm in their sockets. The same method may be pursued with regard to the handle. A little brick-dust mixed with the rosin gives a more secure hold.

Burnishers, contrary to the general opinion of those who manufacture and sell them, should not be too highly polished, but the surface should be rather dull, as in this condition they take a better hold on the gold. If they have too much polish, they glide over the leaf without burnishing it. The polish may be subdued or removed by rubbing the burnishers with crocus martis and oil spread upon a chamois-skin. Care must, of course, be exercised in the operation, or too much of the polish will be removed and the burnisher will require to be repolished. It is, perhaps, needless to say that the burnishers should be kept scrupulously clean and free from everything like moisture and grease.

5.—TIPS. These are made of both camel's hair and badger hair. They are used to lift the gold leaf from the cushion, after it has been suitably cut, and to apply it to the surface prepared to receive it. When they do not freely take up the leaf, it is customary to pass them across the hair of the head, when they receive sufficient electricity from it to attach the gold at once to them.

This attraction has been attributed to the moisture which the tips receive from the hair, but I think the notion is a mistaken one. The tips evidently become electrified, for they will, in that condition, attract the gold at some distance off. Moisture would not produce this effect. The camel's hair tips are the best to use. Several of them will be required. It is a good plan to cut an old one down to within an inch and a half of the paste-board handle, and so use it, when very narrow slips of the leaf are being used; they can be lifted better and more safely with such a contrivance. In all cases, the tip should not be applied to more of the gold to be raised by it than is actually necessary for the purpose.

6.—SOME GLUE, of the best white quality. Many gilders use gelatine instead of glue. For mixing with the burnish-size, the gelatine is undoubtedly the more desirable. The glue should be naturally white, not artificially so-colored; and it should be free from all grit and sediment. Cooper's make, of both glue and gelatine, is undoubtedly the best, for all purposes of gilding. It acts more uniformly and is altogether the most reliable.

7.—PIPE CLAY, for coating the composition parts of the frame. It usually comes in lumps and must be allowed to soak before mixing. The pipe clay has superseded the use of whiting for the two primary white coats to a frame. It is free from all grit, whilst whiting is not, and it gives a soft, smooth surface for the layer of gold leaf.

8.—PARIS WHITE, or whiting. This is used for stopping up holes and other imperfections in the frame. It has more body than the pipe clay, and for that reason is used instead of it. Paris white is simply whiting precipitated in powder, and it is consequently freer from grit and other impurities.

9.—OIL-SIZE. This is used for giving the adhesive coat to which the gold is applied when laid in oil.

10.—BURNISH-SIZE. Used for giving the coat to which the gold is applied when a burnished surface is required.

11.—COTTON BATTING, of a good quality. This is used for pressing down the gold leaf when it is being laid in oil.

12.—GOLD LEAF, of the best quality of deep-colored gold. The leaf comes done up in books of twenty-five leaves each, and made into packs of twenty books each.

13.—AN OX-GALL, strained into a bottle, with about two teaspoonfuls of carbolic acid to preserve it from decomposition. This, mixed with the white coat and the clear size, will prevent pin-holes and frothing. Some gilders use alcohol and even spirits of turpentine for this purpose; vinegar has also been recommended, but, after a thorough trial of all of them, I feel confident that the best results will be had with the ox-gall.

14.—CARBOLIC ACID. The crystallized form is the best to use. It comes in bottles of a pound weight. It must be melted by placing the bottle in hot water; and, when the fluid condition takes place, add some water, when the whole mass will remain fluid. This is a perfect antiseptic and very useful to mix with the glue and gelatine, which are thereby preserved from decomposition.

15.—SAND PAPER; numbers 0, $\frac{1}{2}$ and 1.

16.—ALCOHOL, of the highest proof. This is used in laying the gold leaf which is to be burnished.

17.—JAPAN gold size.

18.—SHELLAC VARNISH, both the brown and the white.

19.—BURNISH-SIZE STRAINER.

20.—SEVERAL CUPS, for mixing the different preparations. It is well to have covers for them, as all the preparations should be effectually protected from dust.

21.—A GRADUATED GLASS MEASURE; one of four ounces capacity is the most useful.

22.—SPOONS, for mixing with. The tea size is the best for general use.

II.

The burnish-size should be always kept in a china or earthenware jar. If kept in a tin vessel, its moisture causes the tin to rust, and the deposit caused by the rust is highly injurious to the size. The oil-size may be kept in a tin vessel provided with a cover. When this size is required for use, it should be thoroughly stirred up, as the coloring matter of which it is in part composed, being the heaviest, has a tendency to settle and to leave the pure oil on the surface. Take out only a sufficient quantity for immediate use, as it sets quite rapidly, particularly in warm weather; and do not return any that may be left, back to the original vessel containing the size. Pay no attention to the skin which from time to time forms on the surface, when not in use.

III.

We will now proceed in earnest to the serious work of gilding a picture frame.

1.—Commence by weighing out one ounce and a half of glue or gelatine. Put this into a cup and pour over it eight ounces of water by measurement. This of course you will do with the aid of your graduated glass measure. Let the glue soak for several hours, and then set the cup in a pan of water and expose it to heat until the glue becomes perfectly fluid and completely melted. A teaspoonful of carbolic acid must now be stirred into it and thoroughly incorporated with it. The acid will preserve it, as already stated, from decomposition, and most effectually so and for any length of time. This glue should be strained before being used. Whenever a portion of it will be required for the preparations herein described, it is well to take out as much only as may be wanted for use, leaving the balance untouched; as by so doing its purity can be relied upon, and besides, the oftener glue is melted the weaker it becomes.

2.—Lay down the frame which is to be gilded, back up, and, with a pencil, mark a line across the sections, if there be any, so as to be enabled to return them to their correct positions when ready to do so. Next, draw out the nails which fasten the sections together, and then take them apart. Now turn the sections face up and proceed to fill up any nail-holes about the corners, where the sides have been joined together, as well as all imperfections which may be found to exist. There are often fractures and indentations to be found, and these must be carefully repaired and the frame, or its several parts, made as perfect as can be. The joining at the corner must be filled up and made to appear solid.

3.—The preparation with which all this is to be done is made as follows:

Mix some of the Paris white with just sufficient water to moisten it thoroughly, and then add enough of the glue which has been described, and which has been melted for the purpose, to make the mixture into a thick paste. The Paris white might be mixed at once with the glue, but it does not seem to combine so well as when it has been previously moistened with water. It will be too lumpy and full of undissolved particles, unless mixed

with the fingers; which operation would be generally objectionable. The paste should be just thick enough not to run.

4.—Now proceed to fill up the nail-holes, corner joints and all other visible imperfections with this mixture; either by applying it with a small bristle brush and laying on a sufficient quantity, until the required places are fully stopped up, and, when perfectly dry, sand-papering them down smooth and uniform, or by applying it like putty, with the fingers. The latter method is not recommended, as it requires a peculiar facility in the management, which can only be acquired by considerable practice, and besides, it does not give as good results.

5.—After smoothing down the fillings-up, the flats and hollows of the frame should be gone over first with number 0 sand-paper, and afterwards with a moistened rag, and be rubbed perfectly smooth and even and until the surface appears polished.

6.—The frame will now be ready for the white coat, which must be prepared in the following manner and then applied to the composition parts only:

Soak some of the pipe clay in water, barely enough of the latter to moisten the clay thoroughly, and then add enough of the glue, already prepared and in so hot a state as to render it of the consistency of thin cream; and add also a teaspoonful of the ox-gall, to prevent pin-holes forming and to keep it from frothing. A little practice will be required to determine invariably the requisite consistency, but the above directions will give a satisfactory result. The object of these two preliminary white coats is to kill any grease which may still adhere to the moulding, and which would otherwise prevent the proper drying of the oil-size. They also serve to give a soft and smooth appearance to these parts. The mixture as described must be applied warm, with a bristle brush (number 4 will do for the purpose), to all the composition parts of the frame. Apply this mixture sparingly, but sufficiently to cover the parts completely, and be careful to avoid bubbles while applying it. When the first coat is perfectly dry, apply the second.

7.—When the last white coat is perfectly dry, two coats of clear size should be applied all over these portions which are to receive a coat of oil-size. Even where a portion of the composition is to be burnished, (which process is done in water,) it is better to go all over these parts with the clear

size; but the hollows and flats which should be burnished must not be touched, if it be possible to avoid doing so.

8.—The clear size is prepared as follows: Take one part of the melted glue, as already prepared, and mix it with two parts of water, adding a little of the ox-gall for the purpose already mentioned, and apply this mixture hot, when required, using a bristle brush for the purpose.

9.—The next step will be to prepare, for the purpose, those portions of the frame which require to be burnished. They must be coated with from three to four coats of burnish-size. Those portions of the composition work which are to be burnished should have one or two preliminary coats of Paris white, applied rather thicker than the white coats of pipe clay; and these coats of Paris white are prepared in the same manner as are those of pipe clay. These coats, of course, must only be applied after the portions of the moulding which are to be burnished have been filled up and smoothed down. The filling up is done with thick whiting, and the smoothing down is to be done with a wet rag.

10.—Many gilders use, as a primary coating for the burnish-size, a preparation which is called *yellow clay*. This yellow clay is supplied by dealers, and is to be mixed in just the same way that the burnish-size is. It gives a more elastic bed for the burnishing process; but the same result may be had by giving an extra coat or two of the burnish-size. I do not recommend its use, particularly for beginners. It only serves to complicate a process that is already complicated enough.

11.—The burnish-size, as it is supplied by the dealers, is not in a condition to be at once used, but must be prepared for that purpose in the following manner:

Take some of the crude burnish-size and mix it in a cup with water, add but little water at a time, and mix thoroughly, until the mixture barely drips from the spoon with which it is being mixed. Now add a teaspoonful or two of the hot glue already prepared. On stirring this up, the mass becomes very thick. Continue to add cautiously more glue until the whole begins to thin down, which it will do almost immediately. Perfect success in the burnish work depends almost entirely upon the proper mixing of the burnish-size. The various stages of the process must be invariably observed, just as they have been here described. If the operator exercises ordinary care only, he

will rarely make a failure; though sometimes failure will occur, in spite of all precaution.

This lot of burnish-size may be treated also as a stock to be used from as required. Owing to the carbolic acid which has been mixed with the glue, it will keep from decomposition an indefinite length of time. Without that antiseptic quality imparted to it, it would spoil in a few days.

As now mixed, it will be found rather hard for the final coat or two, and it must be thinned down with thinner glue, or simply with water, for these coats; but for the first two coats it will be about right, and it must be made thin enough to lay smoothly and evenly on the flats and hollows without clogging the brush. When applied to the ornamental parts, it may be applied thicker. This burnish-size must be strained through the strainer. These strainers are procured from the dealers in gilding materials. Of course, the size will not run through the strainer, but it must be worked through by means of a bristle brush. It may as well be said here that all preparations should be strained, so as to avoid any particles of sediment which may be in the materials.

12.—Two or three coats of this prepared burnish-size must now be applied to those parts which are to be burnished, using a round camel's hair brush for the purpose. These coats must be laid on very smoothly and uniformly. The first coat will go on rather crudely, owing to the absorption of the size by the under surface; but the succeeding coats go on more freely. The size should be applied warm, not hot, and the brush used for the purpose should be stirred up frequently, so as to separate the hairs and allow it to work freely. The surplus quantity must be removed from the brush by drawing it across the sides of the cup. The two final coats should be applied in the same manner, but the size used must not be so strong with glue.

When the final coat is dry, it will then be ready for the application of the gold leaf.

13.—The burnishing of a frame is generally done before the oil gilding. The application of the gold leaf is made in the following manner:

With the graduated glass measure, mix one part of alcohol with three parts of water. This solution, which is called *gin-water*, is the medium by which the gold leaf is applied and affixed to those parts which have been coated with the burnish-size. Lay the frame in a convenient position to work

upon it, and have the gilding cushion, the knife, the tips and the gold leaf conveniently placed for use. The customary way of proceeding is to blow from the book on to the cushion as many leaves as it may be convenient to use. Some expertness is required to perform this operation successfully, and I believe that the time required to obtain this expertness is employed to better advantage in removing from the book, with the aid of the knife, a leaf at a time, when it is required to cut one into a number of pieces; or to cut the leaf as it lays in the book, either with the thumb-nail or with the knife. Of course, this will be a matter of choice with the gilder, as to how he will proceed; and he will be altogether governed by the greater facility with which he can work with either method. When the leaf has been cut into the required size for use, lift, by means of the tip, first drawn across the hair of the head, a piece of the gold leaf, and, after wetting thoroughly with the gin-water the portion where it is to be laid, using a camel's hair brush for the purpose, apply the gold quickly.

There seems to be in the gold an occult attraction towards the gin-water; for the leaf is at once drawn to it, and care must be exercised to have the leaf applied at once as nearly right as possible, for where it goes, there it must remain. It must not be touched until dry; although a cyclopædia informs us that the leaf must be pressed down with a camel's hair brush. Such a proceeding would result in anything but a nice state of affairs, as any one will find who might make the experiment.

14.—Never retouch the gold until dry. If the leaf cracks on going on, which it will do in inexperienced hands; do not mind it, but proceed to lay the gold where required. On moistening with the gin-water, be careful not to touch gold already laid, but wet close up to and adjoining it, and let the next piece of leaf lay or lap a little over the first. The moisture runs from the one into the other and makes the junction, when dry, complete.

15.—When the lay is completed and dry, proceed to patch up any cracks and imperfections. This is done by using the number 6 or 8 lettering pencil, and with its long and flexible point, filled with sufficient of the gin-water, wetting the cracks and imperfections one by one and applying pieces of the gold leaf of the required sizes to cover them up completely.

These pieces at once adhere to and join the lay perfectly, and, when they come to be burnished over, never show in the least where they have been

applied, unless the burnish-size is too harsh and hard. In this case, the double layer of gold is brought out.

16.—In about an hour or so, for the flats and hollows, and rather longer for those places where the moisture settles and collects, and consequently remains longer, the lay will be ready to be burnished. The burnisher should be held at an angle, not too perpendicularly, and applied to the gold, finishing as you go along, and burnishing only a small piece at a time. Burnish right over the leaf, just as it has been laid, without brushing off the loose gold leaf.

17.—Burnish over the loose and the firm gold together. Sometimes, and most generally, there will be imperfections in the burnishing when first completed. These may arise from imperfect adhesion of the leaf, or from the leaf rubbing off when the burnish-size has been made too strong with glue, and so the surface has become harsh and hard. These imperfect places may be repaired by simply rubbing them with a wet rag to remove any leaf still adhering, and, when dry, going over them with a thin coat or two of burnish-size and relaying them with gold; but using the least moisture possible, or otherwise a stain will appear around the edges where the moisture has settled. Of course, such places must be reburnished. No moisture should ever be allowed to get on the leaf where another piece of gold leaf is not at once affixed, as otherwise a stain will be made which will mar the uniform purity of the burnish.

18.—Sometimes the surface will chip under the burnisher, particularly about an edge or corner, when too much pressure is applied. This may arise from several causes, but most generally does from the fact that the first and second coats of size have not been of the proper strength, and consequently have a tendency to crumble when the burnisher is applied with more than ordinary force.

19.—In very warm weather, gilders are in the habit of putting a piece of ice in the gin-water. The object of this is not particularly clear. It is claimed for it, however, that it causes the gin-water to lay better on the surface of the burnish-size.

If a chip should occur in a prominent flat, there is no remedy but to wash off the gold leaf with a wet rag, and down to the hard finish, and to re-do the side entirely over. When not too prominent, fill up the chipped place

with Paris white, and when dry, scrape smooth with a penknife, and after giving two or three coats of burnish-size, re-lay the gold and reburnish.

Be careful, when burnishing edges and corners, to bear as lightly as possible with the burnisher, in order to avoid chipping.

20.—Marks in the burnish sometimes appear which seem to be stains, and which branch off in many ways. Such defects are most probably caused by the gin-water being too strong.

21.—The next step will be to apply the oil gold-size to these parts which have been left bare of gold and which have been thoroughly sized with the clear size. It is customary to previously go over with clear size these places which have been touched with burnish-size and which have not received any coating of gold; otherwise the oil-size would be absorbed and would leave no tacky surface to which the gold leaf would adhere. Sufficient of the gold-size should be taken from the can containing it, after first stirring the gold-size up thoroughly, so as to mix it completely with the oil which generally settles on top of it. A piece of tin or glass is the best thing to work from, and the bristle brush with which the size is to be applied should be well rubbed on it from time to time, to distribute the size thoroughly through the brush.

The size must be applied thoroughly to the frame, not too freely, so as to run and settle in hollows, but just enough to cover the surface with a complete coating. Be particular to work the size into all corners and crevices. The brush should not at any time be overcharged with the size. If the size is not evenly distributed over the frame, those places where it settles will simply coat over with a film, leaving the size underneath still fluid, and when the gold comes to be rubbed in with the blender, the film will quickly rub off, and the gold leaf will consequently not adhere.

If any of the oil-size has got upon the burnished parts, which in spite of all care will occur at times, it must be rubbed off with a chamois-skin and if such places are still dingy after the oil gilding, they will admit of considerable polishing with the chamois.

22.—The coat of oil-size must now be allowed to dry. The time required for this purpose depends altogether upon the drying quality of the size. It is generally made to dry in about ten hours and to retain its tacky surface for several days. It is well to allow it to stand for, at least, from twenty to

twenty-four hours; particularly in the winter time, as the more thoroughly dry it is, the brighter will the gilding appear when completed.

When this coating of oil-size has dried sufficiently, it will be ready to have the leaf applied. The leaf, as it is applied, is pressed down gently with a pad of cotton batting. When the frame has been completely covered, take a soft fitch blender and with it distribute the gold thoroughly into corners and depressions. Be careful not to rub too hard, or the gold will be either rubbed off or assume a greasy appearance. If this be all properly done, the gold will have almost as bright and polished an appearance as if it had been burnished.

23.—After dusting the frame well, to remove as much of the dust and particles of leaf as possible, proceed to give these portions only of the frame which have been gilded in oil a coat of finishing-size, which is prepared in the following manner:

Take one part of the melted gelatine or glue, as already prepared, and add to it four parts of water. Now, with a camel's hair brush, apply this finishing size, hot, or nearly so, very sparingly, but very thoroughly. This finishing size may be colored, if desired, so as to deepen the shade of gold; using for that purpose, dragon's blood, gum gamboge or aniline yellow. Only one coat of this size should be applied. This finishing-size serves to give a uniform appearance to the gold, slightly matting the metallic lustre of it, and having the effect also to absorb and remove the loose gold dust and particles still remaining. It also protects the surface from becoming dingy from smoke or other impurities which frequently prevail in rooms.

24.—If it should be desired to render the appearance of the gold still duller or more matted, a very thin coat of the ormolu, which is hereinafter described, should be applied instead of the finishing-size. The more matted the appearance of the surrounding gold, the brighter and more lustrous will be the burnished part. This is altogether a matter of taste and fashion, sometimes a preference for one effect and sometimes for another, prevailing with the public.

25.—Before the application of the finishing size, any imperfectly covered places must be regilded, using for a size the Japan gold size. This size sets in about an hour, and for that reason is preferable to the ordinary oil-size. The places to be regilt should be first gone over with shellac varnish.

Whenever gold is to be regilt, it must be gone over with shellac, as otherwise it will have a greasy appearance.

26.—Old frames that are to be regilt, should be first washed clean and free from dirt, and then be given a coat of shellac before the oil-size is applied. Where any burnish has been, it must, of course, be washed off before the reburnishing can be done.

Some gilders use shellac varnish in place of the clear size, to size the frame for the oil gilding. It works well enough on the composition parts, but on the flats and hollows it dries so rapidly that it leaves ridges and inequalities to such an extent that the smoothness of the work is spoiled. For quick work, however, and by using the shellac very sparingly and giving two coats of it, very good results are to be had.

27.—Flats are generally matted. Occupying, as they do, a prominent and conspicuous position, being next to the picture which is to occupy the frame, they require to be very perfectly done, and they are therefore done in *water-gilt*, and subsequently matted with a coat and sometimes with two of *ormolu*. The flat to be matted is proceeded with as follows:

After all imperfections have been removed, it must be rubbed smooth with very fine sand paper and polished with a moist rag. Three coats of burnish-size are then applied very evenly and smoothly to it, avoiding all marks of the brush where possible; or two coats of yellow clay, prepared in the same manner as the burnish-size, and afterwards two coats of burnish-size, are given. When the final coat is sufficiently dry, the flat must be polished with a piece of coarse paper, or it may even be burnished, when time is no object. If the inside edge should be intended to be burnished after being laid in gold, that part must not be polished, nor receive the coat of very thin glue water, which is to be applied after the polishing process to all that part which is to be matted. This coat of glue water must be applied very evenly and freely to the flat, avoiding bubbles if possible. When dry, the flat is all laid in gold in the same manner as proceeded with when laying the gold for the burnish work.

28.—The best plan to lay a perfect flat is to double-gild it, which consists in simply laying on another coat of gold after the first has dried. The first coat when dry, as well as the second, must be rubbed smooth with a wad of cotton batting. Care must be exercised during this process, in order to avoid,

as much as possible, disturbing the lay. To lay a second coat over the first, the gin-water must be washed over at once, so as to avoid washing up the gold leaf.

29.—After the gold has been rubbed smooth with the cotton, a coat of ormolu must be applied to the unburnished part, to impart to it the matted appearance required. The ormolu should be applied warm and by means of a camel's hair brush, and the application must be very nicely and carefully done.

30.—The following is the manner in which the ormolu is prepared:

Take a teaspoonful of either the tincture of gum benzoin or of white shellac varnish, and mix it in a cup with about twice or three times its bulk of the highest proof alcohol. Now have ready a hot solution of glue or of gelatine, of about the same strength as that of the finishing size. The quantity should be about one-half a teacupful. This must be poured quickly and all at once into the solution of the gum benzoin or of shellac, and the result will be a perfect emulsion of the gum, which will be of a milky white appearance. This, of course, should be strained, and is to be applied as already described. If the first coat should appear spotty and streaky, a second one will be necessary. In applying either the finishing size or the ormolu, it is well to have a lump of alum convenient, to which, from time to time, the brush may be applied. This will, in the case of either, cause it to go on in a better manner and to lay more evenly.

The ormolu, as prepared by the above method, will keep for any length of time. It may also be slightly colored, if it be desired to give the gold a deeper tinge.

31.—The frame will now be about finished, as far as the gilding is concerned. The sections must now be returned to their respective places and securely nailed together.

The outside edge of the frame, if it has been burnished, will require no further preparation, but if it has been gilded in oil, or if it has simply been laid in water, it should have a coat of white spirit varnish applied with a camel's hair brush. This coating of varnish will render the layer of leaf sufficiently hard to be handled without injury to it.

32.—When the edge has not been gilded at all, it is the practice to give it two coats of yellow ochre, mixed about as strong as those of the pipe clay

and applied with a bristle brush. The yellow ochre should be strained before being applied.

33.—Before anything has been done to the frame, and after the sections have been taken apart, it is well to protect the outside corners with pieces of wood securely fastened to the back. This will prevent them from being injured while being gilded. Gilders generally suspend a frame when not being worked upon, to some projection, to keep it out of the way of injury; but, with the corners protected, it may be stood anywhere with safety.

34.—In the foregoing description of the process of gilding, the burnished part has been described as being done before the oil gilding. The oil gilding may, however, be done first and the burnishing afterwards. This manner of proceeding is generally practiced in Europe, but not so much in this country.

However, when much burnishing is to be done, this method is recommended. All those parts of the frame which are to be oil gilded must be coated with the oil gold-size; and wherever this size has got upon places to be burnished, it must be scrupulously wiped off. Then, after the size has set and before it has been gilded, all these parts which are to be burnished must be gone over with a coat or two of pipe clay, to which has been added an extra quantity of ox-gall, in order to cause these coats to lay perfectly over any spots which the oil-size may have touched. Then apply the burnish-size, as previously described. When burnishing, be very careful not to touch with the burnisher those parts which have been oil gilded, and do not burnish too close to them.

35.—Sanded work is used very much in combination with composition on picture-frames. The sand, or crushed quartz, is made to adhere by means of glue. Such sanded work must have a very thorough coating of Paris white before being gilded.

36.—Plain wood, where the grain is to be shown after being gilt, must be well treated with shellac before having the size applied.

THE BRONZING PROCESS.

I.

The implements and materials required for the bronzing process are the same as those required for gilding, with the exception of the gold leaf, the cushion knife and tips; and with the addition of the BRONZE POWDERS of the necessary colors, and of COPAL VARNISH. These may be obtained from any dealer in art materials; but, of course, those of the best quality only should be used.

II.

1.—The frame which is to be bronzed must be proceeded with in the same manner as described in the gilding process. That is to say, all imperfections must be first remedied and the frame put in perfect order.

2.—Two coats of pipe clay are then to be applied to all of the composition work, using a bristle brush for that purpose. The pipe clay is to be prepared as already described for the gilding process.

3.—Two coats of burnish-size must next be applied all over the frame, using a bristle brush instead of a camel's hair brush on the composition parts. For the smooth parts of the frame, a camel's hair brush may be used, as a more even surface may be obtained in that way. The burnish-size should be applied very sparingly, so as not to clog up the ornamental parts of the frame. After having given the frame two thorough coats of the burnish-size, a third and final coat of the same size must be applied. To this final coat, before being applied, should be added a small quantity of powdered lamp black. The addition of the lamp black is not absolutely necessary, however, as the object for which it is used is simply to give a darker appearance to the final coat, which, in the crevices and hollows, when the whole work receives the varnish coat, then assumes that antique appearance. The lamp black rather impairs the burnishing quality of the size, though not to any serious extent.

4.—After the final coat has been applied, put some of the bronze powder which is to be used into a small saucer. A very little bronze powder will go a great way. In another saucer, pour a weak solution of gelatine or of gum arabic. This must be just sufficiently strong to hold the bronze powder on the frame. The usual way of proceeding, in order to apply the bronze powder, is to dip the fingers first into the gelatine or gum arabic, and then into the bronze powder, and proceed to rub it on to the work to be bronzed. It may be applied with a bristle brush, and, where the powder is to be inserted into cracks and crevices, it must be so applied. The object of using the fingers is, that with them a smoother and thinner coat can be obtained; and the smoother and thinner the coat, the better will it burnish.

5.—When the frame has been completely covered and the coating is sufficiently dry, all the flat and smooth parts of the frame, and all those parts of the composition work also which are in relief, must be completely burnished. This part of the work requires a great deal of patience and time; but one is fully repaid for the labor bestowed, in the brilliancy of the work when done.

6.—After the frame has been burnished sufficiently, a thin coat of copal varnish is to be given all over it. The varnish is intended to protect the bronze from tarnishing through contact with the air, as well as from any moisture which might accidentally get on the frame, and which, of course, would remove the film of bronze.



Transcriber's Notes

- Missing or obscured punctuation was silently corrected.
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